



No guarantees, no trade: How banks affect export patterns[☆]



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ARTICLE INFO

Article history:

Received 26 November 2016
Received in revised form 7 July 2017
Accepted 17 July 2017
Available online 21 July 2017

JEL classification:

F21
F23
F34
G21

Keywords:

Trade finance
Global banks
Letter of credit
Exports
Financial shocks

ABSTRACT

Employing new data on U.S. banks' trade-finance claims by country, this paper estimates the effect of letter-of-credit supply shocks on U.S. exports. We show that a one-standard deviation negative shock to a country's letter-of-credit supply reduces U.S. exports to that country by 1.5 percentage points. This effect is driven by countries that are small and where few banks are active. It more than doubles during the 2007–09 crisis. The provision of letters of credit is highly concentrated and banks are geographically specialized. Therefore, shocks to individual banks can have sizable effects in the aggregate and affect trade patterns.

Published by Elsevier B.V.

1. Introduction

Trading across borders exposes firms to substantial risks. To mitigate these risks, exporters often settle their transactions with letters of credit provided by banks, in particular when selling to risky

destinations or when engaging with new trading partners.¹ What happens when these instruments are not available—for example, because banks have limited balance sheet capacity—is not clear. In principle, firms can buy and sell without the involvement of banks on pre-payment (cash-in-advance) or post-payment (open account) terms, but transacting on such terms may be too risky.²

In this paper, we investigate whether shocks to the supply of letters of credit affect firm exports. Drawing on new U.S. banking data,

[☆] The authors would especially like to thank Geoffrey Barnes and Tyler Bodine-Smith for excellent research assistance. For their helpful comments, they would also like to thank Mary Amiri, Pol Antras, Andrew Bernard, Gabriel Chodorow-Reich, Stijn Claessens, Giancarlo Corsetti, Pablo D'Erasmus, Martin Goetz, Kalina Manova, Atif Mian, Daniel Paravisini, Steve Redding, Peter Schott, Philipp Schnabl, Mine Z. Senses, David Weinstein, and members of the D.C. trade group as well as participants in seminars at New York Fed, San Francisco Fed, Federal Reserve Board, LMU Munich, Ifo Institute Munich, Bank of England, University of Oxford, University of Passau, University of Cambridge, Graduate Institute, LSE, and the following conferences: NBER SI International Macro & Finance, SED 2015, AEA 2015, Econometric Society World Congress 2015, FIRS 2014, RMET 2014 and CESifo Conference on Macro, Money and International Finance.

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¹ The views in this paper are solely the responsibility of the author(s) and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of any other person associated with the Federal Reserve System.

¹ A letter of credit works as follows: The importer asks a bank in her country to issue a letter of credit. This letter is sent to the exporter. It guarantees that the issuing bank will pay the agreed contract value to the exporter if a set of conditions is fulfilled. For example, the issuing bank may promise to pay upon receipt of shipping documents. In addition, a bank in the exporter's country typically confirms the letter of credit, whereby the confirming bank commits to paying if the issuing bank defaults. Niepmann and Schmidt-Eisenlohr (2017) document that letters of credit are used for 8% of U.S. exports (\$116 billion in terms of export values). Antràs and Foley (2015) show that these instruments are employed more often in new trade relationships.

² Trade credit insurance, for example, is not a good substitute for a letter of credit. While insurance shifts risk from the exporter to the insurer, a letter of credit reduces the real risk in the economy by providing a commitment device for the importer. For very risky countries, trade credit insurance is therefore often very costly or even unavailable.

we find that there is indeed a causal effect. A one-standard deviation shock to a country's letters-of-credit supply by U.S. banks reduces U.S. export growth to that country by 1.5 percentage points. This effect is only present for exports to the smaller, riskier destinations and more than doubles during times of crisis, suggesting that constraints in the supply of these instruments likely played a magnifying role during the Great Trade Collapse in 2008–09.

The key challenge in identifying the effect of letter-of-credit supply constraints on exports lies in disentangling supply and demand factors. Data from FFIEC 009 Country Exposure Reports available at the Federal Reserve are helpful in this regard. They contain information on U.S. banks' trade-finance claims, which reflect mostly letters of credit in support of U.S. exports, by destination country at a quarterly frequency over a period of 15 years, and allow us to construct country-level letter-of-credit supply shocks that are independent of demand factors.³

In a first step, we regress growth of bank *b*'s trade finance claims in country *i* at time *t* on bank-time fixed effects and country-time fixed effects. The country-time fixed effects control for time-varying demand factors so that the estimated bank-time fixed effects correspond to idiosyncratic shocks to the bank-level supply of letters of credit. These are positively correlated with growth in loans and negatively correlated with banks' credit default swap spreads, which indicates that the shocks capture idiosyncrasies in banks' business conditions. However, they also pick up strategic decisions by bank managers to expand or contract the letter-of-credit business.

In a second step, we exploit variation in the importance of U.S. banks in the provision of letters of credit across export destinations. The more important bank *b* is for the provision of letters of credit for destination *i*, the larger should be the effect of a change in the supply of bank *b* on exports to destination *i*. Accordingly, we sum the bank-level shocks over all banks that provide letters of credit for destination *i*, weighing shocks by the banks' lagged market shares. In a third step, we regress U.S. export growth to destination *i* at time *t* on the resulting country-level shocks.

If firms cannot easily switch to other banks, because of information asymmetries and the small number of banks active in this business, and trading becomes too risky without a letter of credit, then we should find that our constructed letter-of-credit supply shocks predict U.S. exports. Indeed, we document statistically and economically significant effects. A country-level shock of one standard deviation decreases export growth to that country, on average, by 1.5 percentage points. Investigating the quantitative implications of our regressions further, we find that because of the high concentration of the business, shocks to the supply of a single bank are big enough to have effects in the aggregate. A reduction in the letter-of-credit supply by a large U.S. bank of a magnitude that corresponds to the 10th percentile of the bank-level shock distribution leads to a 0.7-percentage-point decline in total U.S. exports growth.⁴ A shock of the same magnitude to the supply of a different large bank would have different effects across countries since banks specialize, serving different markets to varying degrees. In this sense, banks affect not only export volumes but also export patterns.

Another key finding of this study is that the effects of letter-of-credit supply shocks are heterogeneous across export destinations and over time. Only exports to destinations that are smaller and where fewer U.S. banks are active decline when there is lower supply. The effects of reductions are also stronger during times of financial distress. In a crisis period, the effect of letter-of-credit supply shocks

more than doubles compared to normal times with a beta coefficient of 10%. These findings can be explained as follows. Firms use letters of credit more intensively and are less willing to trade without them when exporting to riskier markets and when economic uncertainty is high.⁵ At the same time, it is more difficult for firms to switch to other banks. There are fewer banks active in smaller markets. Moreover, banks may be less willing to expand to new markets and less able to obtain liquidity or to take on more risk during a financial crisis.

The literature on the link between finance and trade is active and growing. The contribution of our paper is to highlight the effects of financial shocks on trade through, what we call, the risk channel, meaning the role that banks play in reducing risk for exporters. Specifically, we provide the first evidence that the effects of letter-of-credit supply shocks are economically significant, heterogeneous across countries and stronger during times of financial distress. Our research is most closely related to [Ahn \(2013\)](#), who studies the effect of bank balance-sheet shocks on the provision of letters of credit in 2008–2009 in Colombia. His paper nicely complements our analysis. He finds that bank balance-sheet items predict the variation in bank-level letter-of-credit supply during a financial crisis in a small open economy (Colombia), in contrast to the focus on a large country with a sophisticated financial sector (the U.S.) both during normal and crisis times here.

A few additional papers investigate the role of insurers and banks in reducing export risk. [Van der Veer \(2015\)](#) studies trade credit insurance, finding a relationship between the supply of insurance by one large insurer and aggregate trade flows. [Auboin and Engemann \(2014\)](#) exploit data on export insurance from the Berne Union to analyze the effect of insurance on trade. [Hale et al. \(2013\)](#) document that an increase in bank linkages between countries is associated with larger bilateral exports, conjecturing that banks mitigate export risk. [Demir et al. \(forthcoming\)](#) show that the changes to capital weights in Turkey under Basel II affected letter-of-credit based trade.⁶

Other studies in the literature have focused on the effect of shocks to the supply of credit on firm exports, a channel we term the working capital channel. To finance the time gap between production and sales revenues, firms need funding. Because international trade takes longer and, therefore, financing needs are higher, international trade may suffer—more than domestic sales—when credit is tight. Evidence for this is presented in a number of studies. Using Japanese matched bank-firm data, [Amiti and Weinstein \(2011\)](#) show that if a bank has a negative shock to its market-to-book value, a firm that lists this bank as its main bank has a drop in exports that is larger than the observed drop in domestic sales. [Paravisini et al. \(2015\)](#) use matched bank-firm data from Peru to estimate the elasticity of exports with respect to credit. Exploiting heterogeneity in the exposure of banks to foreign funding shocks, they find that credit supply shocks reduced exports during the recent financial crisis. [Del Prete and Federico \(2014\)](#) employ Italian matched bank-firm data with information on export- and import-related lending, showing that Italian banks reduced mainly their supply of general loans and that this lowered Italian exports during the Great Recession. Using the same data source from Italy, [Buono and Formai \(2016\)](#) find that credit shocks affect exports more than domestic sales in normal times.

The paper is also related to the literature on financial development and comparative advantage ([Beck, 2003](#), [Manova, 2013](#)) and, more broadly, to recent research that documents the real effects of financial shocks for employment ([Chodorow-Reich, 2014](#)), firm

³ Our identification strategy follows in large parts [Amiti and Weinstein \(forthcoming\)](#) and [Greenstone et al. \(2014\)](#) although we refine their methodology as discussed in more detail in the conclusions.

⁴ The estimated lower bound of this effect is 0.1 percentage point, the upper bound 1.3 percentage points.

⁵ We show that banks' trade finance claims peaked during the 2007–09 crisis in line with findings based on SWIFT traffic in [Niepmann and Schmidt-Eisenlohr \(2017\)](#).

⁶ The Turkish data has also been used by [Demir and Javorcik \(2014\)](#), who show that increased competition makes exporters sell more on open account terms.

liquidity (Khawaja and Mian, 2008) and durable consumption (Mian and Sufi, 2010), among others.⁷ Our results also provide evidence for the propagation of shocks through global banks (Cetorelli and Goldberg, 2012, Kalemli-Ozcan et al., 2013, and Ongena et al., 2015).

The paper is structured as follows. Section 2 and Section 3 give background information on banks' role in trade finance and the data, respectively. Section 4 discusses the empirical strategy. Section 5 presents the results and robustness checks. Section 6 quantifies the aggregate effects of letter-of-credit supply shocks. Section 7 concludes.

2. A primer on trade finance

2.1. The role of banks in facilitating trade

When exporters and importers engage in a trade, they have to agree on who finances the transaction and who bears the risk. Banks help both with financing and with mitigating the risk. The most common payment form in international trade is an open account. In this case, the exporter produces first and the importer pays after receiving the goods. The exporter pre-finances the working capital, either with funds out of her cash flow or through a loan from a bank. And she bears the risk that the importer will not pay after receiving the goods.⁸

To address this commitment problem, banks offer letters of credit. Fig. 1 illustrates how they work. A bank in the importing country issues a letter of credit, which is sent to the exporter. The letter of credit guarantees that the issuing bank will pay the agreed contract value to the exporter if a set of conditions is fulfilled. These conditions typically include delivering a collection of documents to the bank, e.g., shipping documents that confirm the arrival of the goods in the destination country. In most cases, a bank in the exporting country is also involved in the letter-of-credit transaction. Because there is still a risk that the issuing bank will default on its obligation to pay, the exporter can ask a bank in her country to confirm the letter of credit. The confirming bank thereby agrees to pay the exporter if the issuing bank defaults. To the extent that banks can monitor the transaction, the commitment problem is resolved with a letter of credit, since the exporter is paid only after delivering the goods and the importer commits to paying by making her bank issue a letter of credit.⁹

International trade is riskier than domestic sales because contracts are harder to enforce across borders. In addition, less information about the reliability of trading partners may be available. Accordingly, letters of credit are widely used in international trade and are employed to a much smaller extent for domestic sales. Data from the SWIFT Institute on letters of credit show this. In 2012, around 92% of all letters of credit in support of U.S. sales were related to exports and only 8% to domestic activity.¹⁰

2.2. Market structure of the business

The trade-finance business and, in particular, the market for bank guarantees is highly concentrated. In 2012, the top 5 U.S. banks

accounted for 92% of all trade finance claims in the U.S. For comparison, the share of the top 5 banks in U.S. bank assets in that year was around 80%.¹¹ Del Prete and Federico (2014) present details on the market structure for Italy. There, the business is similarly concentrated. Only ten Italian banks extend trade guarantees. Moreover, while most Italian firms borrow from a few financial institutions, the median firm obtains guarantees from only one bank.

The high concentration is likely due to high fixed costs. When banks confirm letters of credit, they need to work with banks abroad and have knowledge of their credit- and trustworthiness. Banks also have to do background checks on their customers to comply with due diligence requirements and anti-money laundering rules before they can engage in any business abroad. They also need to be familiar with the foreign market and the legal environment there. Such knowledge is costly to acquire and not easily transferable.

Due to the presence of information asymmetries, the importance of relationships, and the resulting high concentration of the market, it should be difficult for a firm to switch to another bank when its home bank refuses to confirm or issue a letter of credit.¹² If a letter of credit cannot be obtained, importers and exporters may not be willing to trade or they may reduce the quantity traded. Then a reduction in the supply of letters of credit has an effect on trade.¹³

2.3. Capital costs and survey evidence on trade-finance gaps

Under Basel rules, banks need to set aside capital when confirming letters of credit.¹⁴ Banks active in the trade finance business have argued that these capital requirements discourage them from supplying LCs since trade finance is a relatively low-return business.

Surveys of banks conducted by the International Monetary Fund and the International Chamber of Commerce support the view that the supply of trade finance can constrain international trade. Asmundson et al. (2011) report that 38% of large banks said in July 2009 that they were not able to satisfy all their customer needs and 67% were not confident that they would be able to meet further increases in trade-finance demand in that year. Greater trade-finance constraints may also come from increases in prices. According to the same survey, letter-of-credit prices increased by 28 basis points (bps) over the cost of funds from 2007 q4 to 2008 q4 and by another 23 bps over the cost of funds between 2008 q4 and 2009 q2. Banks also reported that their trade-related lending guidelines changed. Every large bank that tightened its guidelines said that it became more cautious with certain countries. Thus, constraints may differ by destination country. As we will show in the next sections, this survey evidence is consistent with the results presented in this paper.

¹¹ The calculation is based on the sample of banks that reported positive trade finance activities in at least one quarter over our sample period and is for 2012 q1. The actual degree of concentration of the U.S. banking industry is lower since the calculation includes only banks that report the FFIEC 009 and hold positive trade finance claims, and therefore excludes practically all small and medium-sized banks.

¹² Note that there is no alternative method that reduces commitment problems to the same degree. Trade credit insurance, another option for exporters, does not reduce the risk but instead shifts it to another agent, the insurer. When issuing or confirming a letter of credit, banks actively screen documents and manage the conditional payment to the exporter and thereby resolve the commitment problem. Trade credit insurance also implies a guarantee of payment but has no direct effect on the underlying commitment problem. This difference can best be seen in a model with risk-neutral firms as in Schmidt-Eisenlohr (2013). There, firms demand letters of credit to reduce risk but have no reason to buy trade credit insurance that shifts risk to a third party.

¹³ Note that there is an effect on trade even if firms decide to trade without a letter of credit. It follows from revealed preferences that whenever letters of credit are used, other payment forms generate weakly lower profits. Hence, a reduction in the supply of letters of credit can affect both the intensive and the extensive margins of trade. Quantities decline as letter-of-credit costs go up, which represent variable trade costs. If costs become sufficiently large, trade becomes unprofitable.

¹⁴ See BIS (2011).

⁷ See also Chaney (2016), Leibovici (2015) and Kohn et al. (2016) on the effects of financial constraints on trade and Peek and Rosengren (2000), Peek and Rosengren (1997), and Ashcraft (2005) on the real effects of financial shocks.

⁸ Alternatively, the exporter and the importer can agree on cash-in-advance terms. Then, the importer pays the exporter before receiving the goods. In that case, financing is done by the importer who also bears the risk that the exporter may not deliver.

⁹ See Schmidt-Eisenlohr (2013), Antràs and Foley (2015), Ahn (2014) and Hoefele et al. (2016) for more detailed discussions of different payment forms.

¹⁰ Essentially all letters of credit supporting U.S. exports are denominated in U.S. dollars.

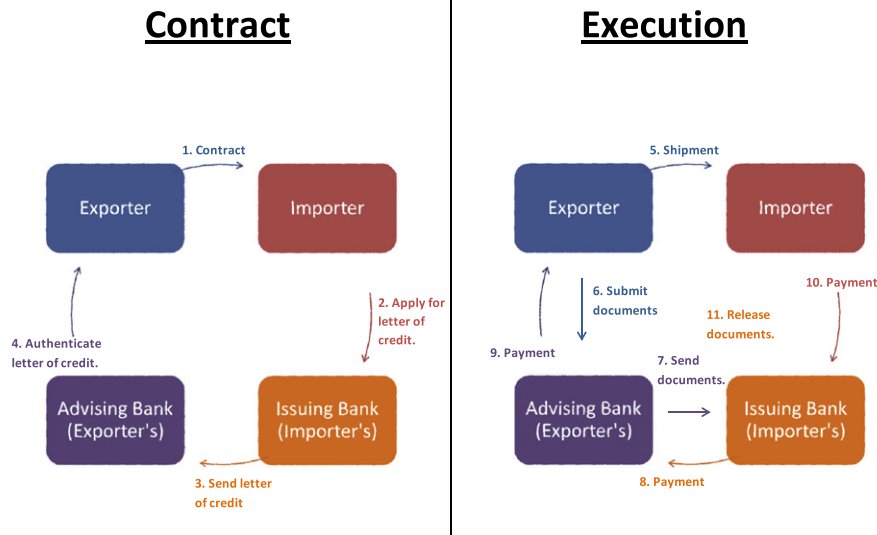


Fig. 1. How a letter of credit works.

3. Data description

The data on letters of credit used in this paper are from the Country Exposure Report (FFIEC 009). U.S. banks and U.S. subsidiaries of foreign banks that have more than \$30 million in total foreign assets are required to file this report and have to provide, country by country, information on their trade-finance-related claims with maturity of one year and under. Claims are reported quarterly on a consolidated basis; that is, they also include the loans and guarantees extended by the foreign affiliates of U.S. banks. The sample covers the period from the first quarter of 1997 to the second quarter of 2012.¹⁵

The statistics are designed to measure the foreign exposures of banks. This information allows regulators to evaluate how U.S. banks would be affected by defaults and crises in foreign countries. Therefore, only information on the claims that U.S. banks have on foreign parties is collected. Loans to U.S. residents and guarantees that back the obligations of U.S. parties are not recorded. While we can rule out based on the reporting instructions that letters of credit in support of U.S. imports or pre-export loans to U.S. exporters are included, it is conceivable that several trade-finance instruments that support either U.S. exports, U.S. imports, or third-country trade constitute the data. We provide a detailed discussion and exploration of the data in online Appendix A, which delivers two key findings. First, banks' trade-finance claims reflect trade finance in support of U.S. exports, and second, the main instrument in the data are letters of credit.

Fig. 2 depicts the evolution of U.S. exports and banks' trade-finance claims over time. Trade-finance claims peaked in 1997/1998 during the Asian crisis and again during the financial crisis in 2007–2009. Since 2010, claims over exports have increased considerably, which is likely due to the low interest rate environment and the retrenchment of European banks from this U.S.-dollar-denominated business, allowing U.S. banks to gain their market shares.

Fig. 3 shows the distribution of trade-finance claims and U.S. exports across world regions in the second quarter of 2012. Regions

are ranked in descending order from the left to the right according to their shares in total trade finance. The upper bar displays the trade-finance shares of the different regions. The lower bar illustrates regions' shares in U.S. exports. While around 50% of U.S. exports go to high income OECD countries, banks' trade-finance claims in these countries only account for around 20%. In contrast, East Asia and the Pacific only receive 11% of U.S. exports, but this region's share in trade finance is twice as large. The figure indicates substantial variation in the extent to which exporters rely on trade finance and letters of credit, specifically, across regions and destination countries, which could lead to asymmetric effects of reductions in their supply. We explore asymmetries in more detail in Section 5.3.

4. Empirical approach

4.1. Estimating letter-of-credit supply shocks

In this section, we discuss the empirical strategy to identify the causal effect of letter-of-credit supply shocks on exports. The

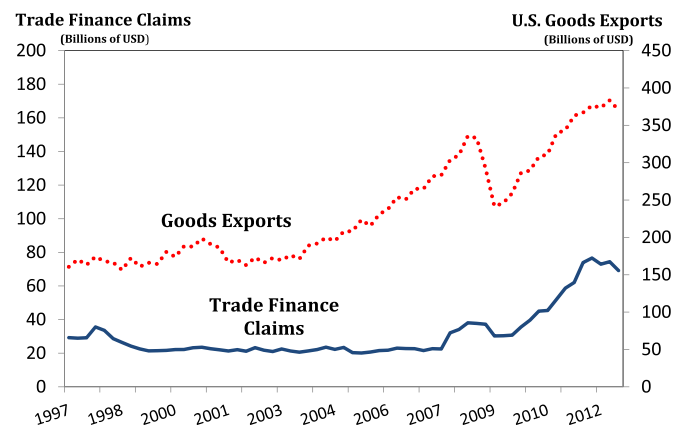


Fig. 2. Evolution of aggregate trade finance claims and U.S. exports over time. Note: The solid line in the graph shows the aggregate trade finance claims of all reporting U.S. banks over time. The years 1997–2000 exclude data from one large bank that changed its trade finance business fundamentally in the reporting period. The dotted line displays the evolution of aggregate U.S. exports in goods over time.

¹⁵ Until 2005, banks' trade-finance claims are reported on an immediate borrower basis; that is, a claim is attributed to the country where the contracting counter-party resides. From 2006 onward, claims are given based on the location of the ultimate guarantor of the claim (ultimate borrower basis). This reporting change does not appear to affect the value of banks' trade-finance claims in a systematic way, so we use the entire time series without explicitly accounting for the change. See <http://www.ffiec.gov/> for more details.

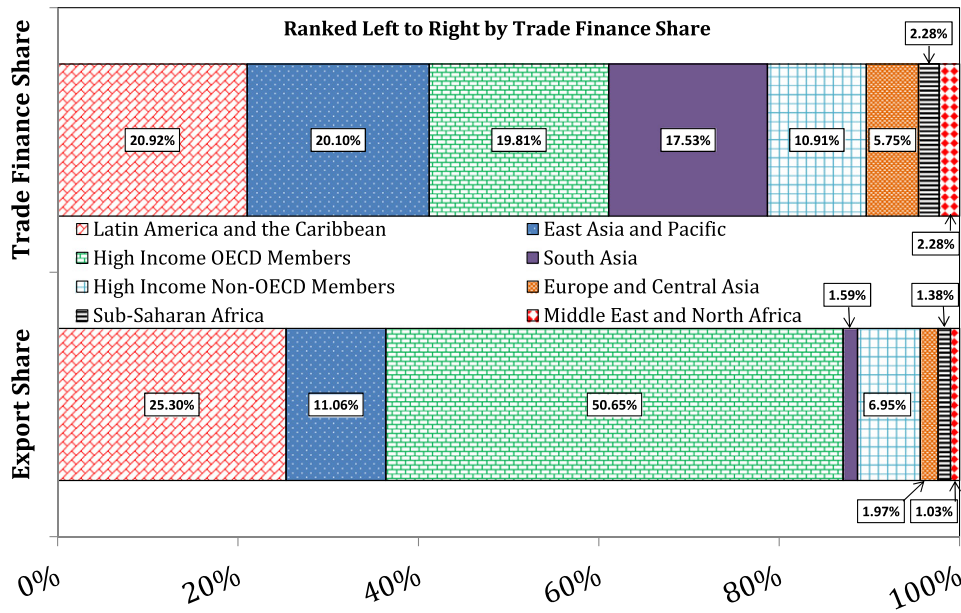


Fig. 3. Trade finance and export shares in 2012 q2 by world region. Note: The upper bar in the graph shows the shares of eight groups of countries in banks' total trade finance claims in 2012 q2. The lower graphs displays each group's share in total U.S. goods exports in that quarter.

challenge in establishing a causal link is to obtain a measure of supply shocks that is exogenous to the demand for letters of credit. Because we have information on the trade finance claims of U.S. banks by destination country that varies over time, we can estimate time-varying idiosyncratic bank-level supply shocks from the data. In line with Greenstone et al. (2014) and Amity and Weinstein (forthcoming), we estimate the following equation:

$$\Delta t f_{bit} = \frac{t f_{bit} - t f_{bit-1}}{t f_{bit-1}} = \alpha_{bt} + \beta_{it} + \epsilon_{bit}, \quad (1)$$

where $t f_{bit}$ corresponds to the trade finance claims of bank b in country i and quarter t .¹⁶ Trade finance claims growth rates are regressed on bank-time fixed effects α_{bt} and on country-time fixed effects β_{it} . If all β_{it} 's were included in the regression together with all α_{bt} 's, the α_{bt} 's and β_{it} 's would be collinear so one fixed effect must be dropped from the regression in each quarter. Without an additional step, the estimated bank-time fixed effect would vary depending on which fixed effect serves as the base category in each period and is omitted from the regression. To avoid this, we regress the estimated bank-time fixed effects on time fixed effects and work with the residuals $\hat{\alpha}_{bt}$ in place of the estimated α_{bt} 's. This normalization sets the mean of $\hat{\alpha}_{bt}$ in each period t to zero and thereby makes it irrelevant which fixed effects are left out when Eq. (1) is estimated.

The obtained bank-time fixed effects $\hat{\alpha}_{bt}$ correspond to idiosyncratic bank shocks. By construction, they are independent of country-time specific factors related to the demand for letters of credit (and, hence, export growth) that affect all banks in the sample in the same way. To further address the concern that bank shocks might pick up demand effects, bank shocks are estimated for each country separately; the bank shock $\hat{\alpha}_{bt}$ used for country i is obtained by estimating Eq. (1) without including observations of country i .¹⁷

The normalized bank-level supply shocks $\hat{\alpha}_{bt}$ are used to construct country-specific supply shocks as follows:

$$\text{shock}_{it} = \sum_b \phi_{bit-2} \hat{\alpha}_{bt}, \quad (2)$$

where $\phi_{bit-2} = \frac{t f_{bit-2}}{\sum_b t f_{bit-2}}$. Thus, bank supply shocks are weighted by the share of bank b in the total trade finance claims on country i at time $t-2$ and are summed over all banks in the sample. In Section 5.2, we show that results also hold when market shares are lagged by an alternative number of quarters or are averaged over several preceding periods. The resulting variable, shock_{it} , measures shocks to the supply of letters for destination country i at time t .

The effect of letter-of-credit supply shocks on exports is estimated based on the following equation:

$$\Delta X_{it} = \frac{X_{it} - X_{it-1}}{X_{it-1}} = \gamma \text{shock}_{it} + \delta_t + \delta_i + \eta_{it}, \quad (3)$$

where X_{it} denotes U.S. exports to country i at time t . Export growth rates are regressed on the constructed country-level letter-of-credit supply shocks as well as on country fixed effects and time fixed effects. The key coefficient of interest is γ .

Under the assumption that the computed country-level supply shocks are not systematically correlated with unobserved characteristics that vary at the time-country level and are correlated with exports, γ corresponds to the causal effect of letter-of-credit supply shocks on export growth. Expressed in formulas, the identification assumption is: $E\left(\left(\sum_b \phi_{bit-2} \hat{\alpha}_{bt}\right) \eta_{it}\right) = 0$. Given the presented strategy, the assumption is satisfied if two conditions hold. First, the estimated shock to the letter-of-credit supply by bank b is not correlated with changes in the demand for letters of credit. This will hold if shocks to the demand of letters of credit affect all banks symmetrically. Second, banks with positive (negative) shocks to their supply of letters of credit in period t do not sort, at time $t-2$, into markets with positive (negative) deviations from trend export growth in period t . We discuss possible violations of these conditions

¹⁶ We estimate equation 1 with OLS. However, results in this paper are very similar when Eq. (1) is estimated with WLS using lagged trade finance claims as weights.

¹⁷ In Appendix B, we lay out this step more carefully, denoting the bank shock used for country i as $\hat{\alpha}_{bit}$. To keep the exposition simple, we do not introduce this notation in the main text.

in detail in Section 5.2. In particular, the reader might worry that the estimated bank shocks pick up changes in the supply of or demand for U.S. exports. We present a battery of exercises that strongly suggest that our results come in fact from shocks to the supply of letters of credit. We also show that sorting as a driver of our findings can be excluded.

Before turning to the description of the data, we also want to discuss the interpretation of γ . γ captures the effect of a letter-of-credit supply shock to country i on export growth to country i relative to export growth in all other countries. A positive shock to the supply of letters of credit for exports to country i may redirect exports to country i , meaning that exports to country i increase at the expense of exports to other countries. Because of such trade diversion, the estimate of γ should be seen as an upper bound for the direct effect of letter-of-credit supply shocks on exports to country i .

4.2. Description of the sample

U.S. banks have trade finance claims in practically all countries of the world but only a few out of all banks that file the FFIEC009 report have positive values. For example, in the first quarter of 2012, 18 banks had positive trade finance claims in at least one country whereas 51 banks reported none. Three banks had positive trade finance claims in more than 70 countries while seven banks were active in less than five countries. Over the sample period, banks drop in and out of the dataset and acquire other banks. To account for acquisitions, the trade finance claims growth rates are calculated in the period of an acquisition based on the sum of the trade finance claims of the acquired bank and the acquiring bank in the previous period. The same adjustment is made when the bank shares ϕ_{bit-2} are calculated. If a bank acquired another bank at time t or $t-1$ we use the country share of the two banks added up to compute bank shares.

Bank supply shocks are estimated on a sample in which observations of claims growth are dropped for which tf_{bit} or tf_{bit-1} is zero.¹⁸ The total claims associated with these observations is small, adding up to a little more than 1% of the value of total claims in the data. We also drop the first and 99th percentiles of the claims growth rate distribution based on the remaining observations, further mitigating the influence of outliers. The dataset used to estimate Eq. (1) and obtain the bank-time fixed effects has 32,256 observations, covers the period from 1997 q2 to 2012 q2 and includes 107 banks as well 159 countries (for summary statistics, see Table 1).

4.3. Exploring the estimated bank-level shocks

In total, we estimate 325,389 time-country-varying bank shocks.¹⁹ The number of banks goes down over time due to consolidation in the banking sector. In the third quarter of 1997, we obtain bank shocks for 54 different banks, down to 18 banks in the second quarter of 2012. Fig. G.1 shows the distribution of bank shocks, which exhibits significant variation. Table 1 provides the corresponding summary statistics. Fig. G.2 displays the median normalized bank shock and the standard deviation of the bank shocks over time. Note that the mean is by construction equal to zero in each quarter as shown in Table 1.²⁰

Table 1
Summary statistics.

Variable	(1) N	(2) Mean	(3) SD	(4) Min	(5) Max
Trade finance growth Δtf_{bit}	32,256	0.225	1.049	−0.904	9.8
Bank shock $\hat{\alpha}_{bit}$	325,389	0	0.500	−2.316	5.806
Country-level shock ϵ_{it}	6751	0.030	0.171	−0.97	2.202
Country-level shock ϵ_{it} in sample with controls	4949	0.032	0.174	−0.971	1.813
Export growth ΔX_{it} in sample with controls	4949	0.057	0.309	−0.667	2.06

Note: In the first row, summary statistics of trade finance growth rates are given that are observed in the sample that is used to estimate Eq. (1). The second row provides summary statistics of the normalized bank shocks that are obtained from estimating Eq. (1), always dropping country i from the sample. The summary statistics of the country-level shocks in the third row are for all country-level shocks that are computed. In the fourth row only those country-level shocks are included that are used in the estimation of equation 7 with controls. The last row displays summary statistics of the corresponding export growth rates.

We check whether bank shocks are correlated with meaningful bank-level variables.²¹ Banks allocate funding to business lines and may cut funding as overall conditions worsen. Because letters of credit are short term and contracts are liquidated within a few months or even weeks, letter-of-credit supply can be quickly reduced to shrink banks' overall balance sheet, reduce exposures and improve liquidity. However, banks may also take strategic decisions to grow or contract letter-of-credit supply for other reasons. Thus, the estimated bank-level shocks may also capture changes in the supply of letters of credit that are not closely linked to the current health of the bank. Banks may, for example, decide to contract their operations with foreign entities to refocus on core activities or when due diligence requirements change.²² There is in fact anecdotal evidence that, due to recently elevated due diligence requirements, some banks have reduced their cooperation with foreign banks. Moreover, European banks refocused their business toward their home countries after the European sovereign debt crisis, also cutting down on the provision of letters of credit, which allowed U.S. banks to grow. The empirical strategy pursued in this paper allows us to capture changes in banks' supply of letters of credit for all of these reasons.

While we would therefore not expect the balance sheet variables to fully predict the estimated bank-level shocks, it is interesting to understand the extent to which they do. To that end, the bank shock α_{bt} is regressed on deposit growth, loan growth and the credit default swap spread on 6-months senior unsecured debt of bank b at time t . Results are displayed in Table 2. In all columns, bank and time fixed effects are included and standard errors are clustered at the bank level.²³ The results in columns (1) and (2) of Table 2 indicate that the average bank shock is positively correlated with loan growth. Columns (3) and (4) show that it is negatively correlated with banks' credit default swap spreads, an implicit measure of banks' funding costs. The correlations between these bank-level variables and the estimated shocks becomes stronger in the second half of the sample (see columns (2) and (4)).²⁴

¹⁸ For 8.5% of all observations $tf_{bit} = 0$. If $tf_{bit-1} = 0$, claims growth rates in quarter t have to be dropped because they go to infinity. To make the estimation less prone to outliers and keep things symmetric, we also drop negative growth rates of 100%.

¹⁹ Recall the bank-time fixed effects are estimated for 159 different countries, so for each estimation of Eq. (1), we estimate around 2100 bank-time fixed effects.

²⁰ Bank-time fixed effects alone explain 12.1% of the variation in export growth. The R^2 resulting from the estimation of Eq. (1) with bank-time and country-time fixed effects is 34.3%. For more details, see Table H.11.

²¹ Balance-sheet information for banks in the sample comes from the Y-9C and FFIEC 031 reports. Credit default swap spreads are taken from Markit.com.

²² See Working Group on Trade, Debt and Finance (2014) for a summary of recent developments in trade finance after the 2007–2009 crisis.

²³ Note that reverse causality should not be a problem because letters of credit constitute a small fraction of banks' total activities. The median share of trade finance claims in total assets for the banks in our sample is below 1%.

²⁴ We also included measures of profitability in the regressions, e.g. return on equity and return on assets, but the associated coefficients were not significant.

Table 2
Correlation of estimated bank shocks with bank-level variables.

Dep. var $\hat{\alpha}_{bt}$	(1)	(2)	(3)	(4)
	All	After 2006	All	After 2006
Loan growth _{bt}	0.333* (0.184)	0.829*** (0.278)		
Deposit growth _{bt}	0.0196 (0.196)	−0.271 (0.308)		
CDS spread _{bt}			−0.0838** (0.0351)	−0.0862*** (0.0212)
Observations	2179	469	256	150
R ²	0.073	0.069	0.343	0.432

Note: This table analyzes the relationship between the estimated bank shocks and bank-level variables. The dependent variable is the mean bank shock $\hat{\alpha}_{bt}$, which corresponds to the value of $\hat{\alpha}_{bit}$ averaged over all countries. CDS spread is the bank-specific current default swap spread of bank b at time t . All regressions include bank and time fixed effects. Standard errors clustered by bank and are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

4.4. Heterogeneity and persistence in banks' market shares

The empirical strategy in this paper requires that the importance of single banks be heterogeneous across destination markets.²⁵ Otherwise, all countries would be subject to the same shock and we would not be able to identify effects. In addition, it is essential that banks have stable market shares over time, because we use lagged values to compute country shocks. If banks' market shares were very volatile, then lagged values would not contain useful information about the degree to which bank-level supply shocks affect different countries.

The upper panel of Table H.1 shows summary statistics of ϕ_{bit} , the share of bank b in the total trade finance claims of all U.S. banks in country i at time t , at different points in time. There is substantial heterogeneity at every date. The average bank share increased from 2000 until 2012, in line with the observed reduction in the number of banks reporting positive trade finance claims.²⁶ Bank shares range from below 0.1% to 100%. The standard deviation is 27% in the first quarter of 2012.

We check whether bank shares are persistent in two different ways. First, we regress the market share ϕ_{bit} of bank b in country i at time t on country-bank fixed effects. These fixed effects alone explain more than 77% of the variation in bank shares, which implies that there is much cross-sectional variation in banks' market shares but little time variation. Second, we regress the current market share ϕ_{bit} on its lagged values. Without adjusting for mergers and acquisitions, the one-quarter lagged bank share explains around 84% of the variation in the current share, as shown in Table H.2.²⁷ Two-period lagged values, which are used to construct country supply shocks, still explain around 77% of the variation (see the R^2 in column (2)). Column (3) includes bank fixed effects, showing that the high correlation between current and lagged market shares does not come from systematic differences in size across banks.²⁸

Having established strong heterogeneity and persistence in banks' market shares, we compute the country-level supply shocks

shock_{ct} following Eq. (2) for 156 different countries.²⁹ Table 1 displays the summary statistics for this variable.

In the regressions of export growth on country-level supply shocks (Eq. (3)), countries with a population below 250,000, offshore financial centers and observation in the top and bottom one percentile of the export growth rate distribution are excluded from the sample.³⁰ To control for export demand in the destination country, we add a set of variables. This lowers the number of observations further since these variables are not observed for all countries. However, the properties of the country-level supply shocks are unchanged as the summary statistics for the shock variable of the reduced sample show (see again Table 1).

5. Results

5.1. Baseline results

Table 3 presents the baseline regression results obtained from estimating Eq. (3). Unless stated otherwise, standard errors are bootstrapped for all regressions in this section.³¹ In column (1), export growth is regressed on letter-of-credit supply shocks and time fixed effects. The estimated effect of supply shocks is positive and significant at a 1% significance level. The positive coefficient indicates that destination countries that experience larger declines in the supply of letters of credit exhibit lower export growth rates. In column (2), several independent variables that control for changes in import demand are included in the regression: GDP growth, the change in the USD exchange rate of the local currency, and growth in non-U.S. imports of country i in period t . In column (3), country fixed effects are added. The inclusion of the additional variables and fixed effects does not affect the magnitude of the estimated coefficient of interest γ . This shows that letter-of-credit supply shocks are not systematically correlated with time-invariant country characteristics or with demand factors.

Column (4) demonstrates that the identified effect is not driven by simultaneous changes in bank lending. The column includes as an additional control destination-specific loan supply constructed by (i) weighing the loan growth of bank b at time t by ϕ_{bit-2} , the trade finance claims share of bank b in country i at time $t - 2$, and (ii) by summing it over all banks. Even though bank-level loan growth is correlated with bank-level letter-of-credit supply shocks as shown in Table 2, the loan growth control is not significant. This suggests that the effect on export growth is indeed driven by changes in the supply of letters of credit and not loans.

Based on the coefficient of 0.0869 displayed in column (3), a country supply shock of one standard deviation increases export growth to that country by 1.5 percentage points. This corresponds to about 5% of one standard deviation of export growth rates. As a reference, Table 1 provides summary statistics of export growth rates in the sample. We discuss the magnitude of the effect in more detail in Section 6.

To explore which banks are responsible for the effect on exports, we compute supply shocks for the five biggest letter-of-credit suppliers and the remaining banks separately and rerun the baseline regression.³² Column (5) of Table 3 shows the results. The

²⁵ Here, we focus on geographic specialization in banks' letter of credit business. In a more recent paper, Paravisini et al. (2014) find evidence for specialization of banks across markets in their provision of general loans.

²⁶ Changes in banks' market shares over time are slow but substantial. Therefore, we cannot use market shares in the beginning of the sample period and keep them constant over time to obtain country-level shocks.

²⁷ If we adjusted for M&As, then persistence would be even higher.

²⁸ A similar exercise can be conducted for the number of banks n_{it} that are active in a given market i . In Appendix C, we show that similar to market shares, the number of banks in a market is highly persistent.

²⁹ The number of countries reduces slightly because lagged bank shares are not observed for all countries for which we can obtain bank-level shocks.

³⁰ A list of countries designated as offshore financial centers can be found in the Appendix. Banks' trade-finance claims in offshore centers are barely correlated with U.S. exports to these destinations so we drop them since we do not expect a link between trade finance and real activity.

³¹ Clustering at the country level essentially delivers the same standard errors. We bootstrap standard errors to account for the fact that shock_{it} is a generated regressor.

³² We take the five banks with the largest letter-of-credit provision over the sample period and also include merged entities that were separate banks in earlier years.

Table 3
Baseline results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dep. var. ΔX_{it}						1997–2003	2004–2012
Shock _{it}	0.0875*** (0.0333)	0.0864** (0.0340)	0.0869** (0.0375)	0.0875** (0.0360)			
Shock smaller banks _{it}					0.0921* (0.0537)	0.157** (0.0773)	−0.00584 (0.0739)
Shock top 5 banks _{it}					0.0840* (0.0483)	−0.0101 (0.108)	0.102** (0.0505)
Loan growth _{it}				−0.0306 (0.106)			
GDP growth _{it}		−0.0727 (0.0720)	−0.127 (0.0796)	−0.127* (0.0729)	−0.127* (0.0708)	−0.0621 (0.127)	−0.203* (0.111)
USD xrate growth _{it}		−0.212*** (0.0696)	−0.272*** (0.0853)	−0.272*** (0.0666)	−0.272*** (0.0734)	−0.174* (0.102)	−0.510*** (0.164)
Non-U.S. import growth _{it}		0.366*** (0.0511)	0.353*** (0.0548)	0.353*** (0.0543)	0.353*** (0.0555)	0.333*** (0.0751)	0.365*** (0.0794)
Country FE	No	No	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5357	4949	4949	4949	4949	2120	2829
R ²	0.049	0.067	0.100	0.100	0.100	0.111	0.109

Note: This table reports our baseline results on the causal impact of trade finance supply shocks on U.S. export growth. The dependent variable is the growth rate of U.S. exports to country i at time t . $shock_{it}$ is the constructed country-level trade finance supply shock. $loan\ growth_{it}$ is a proxy constructed by (i) weighing the loan growth of bank b at time t by ϕ_{bit-2} , the trade finance claims share of bank b in country i at time $t-2$, and (ii) by summing it over all banks. Column (5) includes country-level shocks that are obtained by aggregating separately the bank-level shocks of the top 5 banks and the remaining banks. Column (6) includes all years prior to 2004. Column (7) includes the years from 2004 onward. All regressions include a constant. Standard errors are bootstrapped and are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

coefficients on the shocks attributed to the top five banks and the remaining banks are both significant and very similar. It may be surprising that small banks can have an effect in the aggregate. However, smaller banks specialize in certain markets so that they can be large and important for the provision of letters of credit for particular destinations. In column (6), the regression with separate shocks for the top five banks is run on a sample that includes years prior to 2004. Column (7) includes all years beginning with 2004. The sample split highlights that banks other than the top five are responsible for the effect on export growth in the early years of the sample, whereas the top five banks drive the effect in the later years. This finding is likely explained by the fact that the market shares of the top five banks steadily rose over the sample period. Since the banking sector went through a prolonged phase of consolidation, the impact of the top five banks on the total supply of letters of credit increased as smaller banks exited and the letter-of-credit business became more concentrated.

5.2. Identification and robustness

In this section, we present several robustness checks. In particular, we address the concern that the constructed country-level shocks could be endogenous to changes in export supply or export demand. We also show that the results are robust to lagging banks' market shares by an alternative number of periods when constructing the country-level supply shocks. Combined with the fact that the estimated bank-level shocks are not positively serially correlated, this rules out endogeneity due to sorting of banks into markets.

5.2.1. Ruling out demand effects

For our identification strategy to work, changes in the demand for trade finance should affect banks symmetrically. This condition would be violated if banks were fully specialized along a certain dimension, e.g. in firms, industries or products, and there was a shock to demand or production of the firm, industry or product. To address this concern, we restrict the set of countries employed when estimating the bank-level supply shocks. First, we only use observations for the large countries in the sample to estimate Eq. (1). Thus, we obtain bank-time fixed effects that exclusively reflect the growth or contraction of letter-of-credit provision by U.S. banks in the large export

destinations, whose imports from the U.S. should be more diversified in terms of firms, industries etc. We use these bank-level shocks to construct country-level supply shocks as before and rerun the regressions on the sample of small and large countries. We define a country to be small if its log exports over the sample period lies below the sample median. Thus, the designation of a country into small or large is constant over time.³³

Table 4 presents the results. When the export equation is estimated based on the sample of large countries, the effect of letter-of-credit supply shocks is close to zero and insignificant (columns (1) and (2)). Thus information on letters of credit supplied to large countries does not predict U.S. export growth in large destinations. If the bank-level shocks picked up changes in the demand for letters of credit, we should see the opposite. The estimated shocks therefore have to either represent letter-of-credit supply shocks or be uninformative but they cannot be driven by demand-side factors. Column (3) shows that the shocks are able to predict export growth in the small destination countries that were excluded in the construction of the supply shocks. The samples of small and large countries do not systematically vary in their ratio of trade finance claims to U.S. exports, so the hypothesis that our results are driven by the demand for letters of credit is hard to defend in consideration of this evidence.

While the first robustness check presented above is, in our view, conclusive in regard of the role of demand effects, we do additional exercises that more specifically address concerns about industry and regional specialization. In the first exercise, we drop all countries from the fixed effects regression whose industrial composition of imports from the U.S. is similar to that of the country for which the supply shocks are constructed. In the second exercise, we drop all countries that are in the same region as country i when estimating Eq. (1). In both cases results remain unchanged. For the details, please see Appendix E.

5.2.2. Evidence against specialization of banks in industries

We also check more directly for evidence that banks specialize in particular firms or industries. More specifically, we test whether

³³ Countries designated as small are listed in the data Appendix.

Table 4
Robustness: Excluding small countries.

Dep. var. ΔX_{it}	(1) Large cntry	(2) Large cntry & crisis	(3) Small cntry	(4) Small cntry & crisis
Shock _{it}	−0.0117 (0.0226)	0.0106 (0.0475)	0.0788* (0.0470)	0.175 (0.127)
GDP growth _{it}	0.00378 (0.0718)	−0.0672 (0.201)	−0.245* (0.131)	−0.209 (0.389)
USD xrate growth _{it}	−0.0776 (0.0631)	−0.129 (0.236)	−0.491*** (0.165)	−0.947 (0.603)
Non-U.S. import growth _{it}	0.381*** (0.0481)	0.446*** (0.0951)	0.309*** (0.0776)	0.371 (0.283)
Observations	2782	382	2561	363
R ²	0.212	0.448	0.074	0.150

Note: The table shows the results when only large export destinations are used to estimate bank-level shocks. In column (1) only large destinations are included in the sample. In column (2), only large countries and quarters from 2007 q3 to 2009 q2 are included. Column (3) is based on a sample with small export destinations only. Columns (4) shows the effect for small export markets and the crisis period. All regressions include a constant, time- and country-fixed effects. Standard errors are bootstrapped and are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

the market shares of particular banks (our ϕ_{bit} 's) correlate with the export shares of particular industries and whether the estimated bank shocks are correlated with specific industry shocks. Appendix F explains in detail these two exercises. Regression results indicate that the market shares of banks do not co-vary systematically with the export shares of particular industries. We also do not find any evidence that industry shocks explain bank shocks.

Beyond the econometric evidence, there are several facts that make it very unlikely that banks specialize in industries or that exports of single firms could drive changes in the letter-of-credit provision of single banks and aggregate export growth rates at the same time. There are only a few banks that provide letters of credit, while there are many more firms and industries. While it is true that international trade is highly concentrated, concentration is much lower than in the letter-of-credit business. Bernard et al. (2009) report that in 2000 the top 1% of exporters, or 2245 firms, were responsible for 80.9% of all U.S. exports. In contrast, in our data in 2012, 5 banks accounted for more than 90% of U.S. banks' trade finance claims. It is unlikely that idiosyncratic shocks to any of these 2245 firms would matter for the business of any of these very large letter-of-credit suppliers. So, the mere fact that the provision of guarantees is concentrated in a few large banks makes specialization improbable. Also, the largest firms are less likely to rely on letters of credit. Larger firms have longer lasting relationships and are better able to cope with risks, since they are big and can diversify within the firm.³⁴ Moreover, a substantial amount of their trade is intra-firm and does not require bank guarantees. Third, banks should seek to spread trade financing over different industries and firms. On one hand, banks want to diversify risks. On the other hand, the costs associated with gathering letter-of-credit-relevant information about a destination and establishing a network of correspondent banks is likely much higher than the cost of acquiring knowledge about an industry.

5.2.3. Sorting

The previous discussion addresses concerns that the idiosyncratic bank shocks we obtain could be endogenous to export growth. Any remaining endogeneity between country-level shocks and export growth rates must thus come through banks' market shares. Recall the identification assumption that banks with positive (negative) shocks in period t do not provide more (less) letters of credit in period $t-2$ to markets with positive (negative) deviations from trend export growth in period t .

In columns (1), (2) and (3) of Table H.4, banks' market shares are lagged by one, three and four quarters, respectively, when computing the country shocks Δt_{it} , in contrast to the two-quarter lags used in the baseline specification. In column (4), four-quarter rolling averages of banks' market shares lagged by one period are used. In column (5), the yearly average market share of each bank is applied to construct the country-level shocks in the next year. The effect of supply shocks on export growth remains significant at a 10% level throughout. Given these results, our identification strategy could be violated only if banks that anticipate growing in period t sort, in period $t-1$, $t-2$, $t-3$ and $t-4$, into markets with higher deviations from trend export growth in period t . As we show in Appendix D, the estimated bank-level shocks are not positively serially correlated. As results hold independent of the number of lags used for the market share variable, systematic sorting of banks period by period can be ruled out.

We also run a Placebo test where we replace U.S. export growth by trade growth of the EU15 countries, as well as additional regressions that include country time trends. To address possible concerns related to our sample selection and the dropping of zeros, we compute trade finance claims growth rates following Davis et al. (2007) to re-estimate Eq. (1) and construct letter-of-credit supply shocks based on the resulting bank-level shocks. The various exercises demonstrate the robustness of our results. Details are reported in Appendix E.

5.3. Heterogeneous effects

In this section, we explore whether effects differ over time and across countries. In Table 5, the sample is split into a crisis, pre-crisis and post-crisis period. The crisis period goes from the third quarter of 2007 to the second quarter of 2009. When the export equation (Eq. (3)) is estimated only on the crisis sample, the point estimate of γ shown in column (1) is significant at a 1% significance level. While the coefficient is not significant for the post-crisis sample in column (3), it is significant for the pre-crisis period in column (2), highlighting that the effect of letter-of-credit supply shocks is also present in normal times (see also Table 3 column (6)).³⁵ The coefficient of 0.178 for the crisis period suggests that a country-level shock of one standard deviation decreases exports by more than 3 percentage points during periods of financial distress, a much larger effect than that obtained based on the full sample (coefficient of 0.0869, column 3 of Table 5). To test formally for differences in the effect over time, we include an interaction term between the shock and a dummy variable for the crisis period in column (4) of Table 5. The coefficient of the interaction term is significant at a 10% level and indicates that the effect is more than double as large during crisis times.

The effect of letter-of-credit supply shocks on export growth does not only vary over time but also across export destinations. In columns (5) and (6) of Table 5, the effect of letter-of-credit supply shocks is estimated based on a sample that only includes small and large export destinations, respectively. Reductions in the supply of letters of credit only have an effect on small export destinations.

³⁴ Antràs and Foley (2015) report that the large U.S. food exporter they study employs letters of credit for only 5.5% of its exports. This is substantially smaller than the 8.8% found by Niepmann and Schmidt-Eisenlohr (2017) for overall U.S. exports. See Monarch and Schmidt-Eisenlohr (2016) for evidence that relationships of larger firms are systematically longer-lasting.

³⁵ The regression in column 6 of Table 3 employs data from the years 1997 to 2003 and hence does not include the crisis of 2007–09.

Table 5
Heterogeneous effects on export growth across countries and over time.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dep. var. ΔX_{it}	Crisis	Pre-crisis	Post-crisis	All	Small cnty	Large cnty	All	Small cnty & crisis	All
Shock _{it}	0.178*** (0.0682)	0.0837* (0.0482)	−0.0846 (0.107)	0.0575 (0.0423)	0.187*** (0.0629)	−0.00195 (0.0266)	0.0411 (0.0348)	0.318** (0.145)	0.00177 (0.0392)
Shock _{it} × crisis dummy _t				0.120* (0.0720)					0.137* (0.0712)
Shock _{it} × small cnty dummy _i							0.0818 (0.0518)		0.0923* (0.0557)
GDP growth _{it}	−0.133 (0.236)	−0.177* (0.0938)	−0.0888 (0.193)	−0.124 (0.0801)	−0.233* (0.134)	−0.0409 (0.0710)	−0.130* (0.0777)	−0.206 (0.418)	−0.126* (0.0756)
USD xrate growth _{it}	−0.583* (0.345)	−0.254*** (0.0847)	−0.0670 (0.282)	−0.273*** (0.0735)	−0.421** (0.197)	−0.163** (0.0720)	−0.271*** (0.0822)	−1.005 (0.624)	−0.272*** (0.0781)
Non-U.S. import growth _{it}	0.492*** (0.171)	0.274*** (0.0573)	0.534*** (0.149)	0.352*** (0.0529)	0.364*** (0.0793)	0.377*** (0.0480)	0.354*** (0.0519)	0.490 (0.326)	0.352*** (0.0524)
Observations	709	3325	915	4949	2248	2701	4949	328	4949
R ²	0.203	0.111	0.116	0.100	0.079	0.228	0.100	0.164	0.101

Note: This table tests for differences in the effect of trade finance supply shocks on U.S. export growth over time and across countries. The dependent variable is the growth rate of U.S. exports to country i at time t . $shock_{it}$ is the constructed country-level trade finance supply shock. Column (1) only includes observations during the crisis period from 2007 q3 to 2009 q2. Column (2) includes observations before the crisis period (before 2007 q3). Column (3) shows results for data after 2009 q2. Column (4) includes a crisis dummy variable takes a value of one during the crisis period. Column (5) is based on a sample of small U.S. export destinations with log U.S. exports below the sample median. Column (6) includes only large export destinations. Column (7) is based on the full sample, where small cnty dummy_i takes a value of 1 if log U.S. exports to a destination are below the sample median. Column (8) includes observations for small export destinations during the crisis period. Column (9) uses the full sample and includes both dummies. All regressions include a constant, time- and country-fixed effects. Standard errors are bootstrapped and are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

While the point estimate of the shock is highly significant and takes a value of 0.187 for small countries in column (5), it is essentially zero and insignificant when only large export destinations are included in the sample in column (6). The difference is confirmed in column (7), in which an interaction term between the shock and a dummy variable for small countries is included, although the coefficient of the interaction is only significant at a 12% level. In a next step, we jointly investigate differences over time and across countries. Column (8) includes only small countries and the recent crisis period. In column (9), the export equation is estimated on the full sample and includes now both the crisis interaction and the market size interaction. These additional regressions show that the effect of letter-of-credit supply shocks on export growth is strongest for exports to small destinations in times of financial distress in the U.S. economy. Then a negative letter-of-credit supply shock of one standard deviation can lead to a reduction in exports of more than 4 percentage points.

To check whether it is really the size of an export market that leads to differences in the effect across countries, we explore alternative sample splits and introduce interaction terms between the shocks and different variables. The evidence suggests that a crucial determinant of the strength of the effect on export growth is the number of U.S. banks that provide letters-of-credit for exports to a given destination country as shown in Table H.6. In column (1), the export regression is estimated only for countries in which less than five U.S. banks are active in quarter $t-1$. Column (2) shows the results for countries with at least five banks and column (3) is based on the full sample and adds interaction terms. The effect of letter-of-credit supply shocks on export growth is clearly larger for countries in which less than five banks are active. The interaction term between a dummy for countries with at least five U.S. banks is large, negative and significant at a 10% level.³⁶

The presented results indicate that if banks contract their supply of letters of credit, only export growth to small countries is affected. This perhaps surprising result can be rationalized. First, only a few

U.S. banks provide letters of credit for small destinations.³⁷ If one of the banks active in those markets reduces its supply, it is especially difficult for trading partners to find an alternative. Second, selling to smaller markets with which trade is less frequent might be riskier. At the same time, trade insurance, which is an alternative to a letter of credit, is more likely to be unavailable or very costly.

Our second finding, that the effect of supply shocks is larger during a crisis period, can also be explained by similar factors. During a period of financial distress, trading partners may find it harder to switch to another bank when the core bank refuses to issue or confirm a letter of credit. Other banks may be less willing to expand their letter-of-credit business to a new market during these times, and banks with existing relationships to intermediaries in a foreign country may not be able to obtain liquidity or may not want to add risk to their balance sheets. At the same time, exporters and importer may be more reluctant to trade without a letter of credit as they are more risk averse. Direct evidence for this is presented in Niepmann and Schmidt-Eisenlohr (2017) who find that the use of letters of credit relative to exports increased during the 2007–2009 crisis.

5.4. Dynamic effects

Are the identified effects of letter-of-credit supply shocks long-term or transient? To answer these questions, we include lags and leads of the shock variable in the regressions. Results are reported in Table 6. We find that the one-quarter lagged letter-of-credit supply shock also explains export growth. Higher order lags and leads are not significant. The negative coefficient of the one-quarter lagged shock suggests that reductions in export growth from a letter-of-credit supply shock in period t are partly offset by higher export growth in period $t+1$. To better understand this result, we run regressions separately for the crisis sample and the non-crisis sample as shown in columns (4) and (5). While the effects of letter-of-credit supply shocks are in large parts transient in normal times, they are more persistent in crisis times indicated by the much larger coefficient on the contemporaneous shock (0.167) than on the lagged shock (−0.0645) in column (4). Reductions in the supply of

³⁶ We also study the role of market concentration. Specifically, we use the banks' trade finance market shares to compute a Herfindahl index for each country and run regressions including a dummy for above average concentration. We find that the effect of letter-of-credit supply shocks is higher for countries with higher market concentration.

³⁷ Evidence on this is and on how other country characteristics affect the number of active U.S. banks can be found in Table 6 of the working paper version.

Table 6
Dynamics.

	(1)	(2) All	(3)	(4) Crisis	(5) No crisis
Shock _{it+1}			−0.00273 (0.0329)		
Shock _{it}	0.0892** (0.0350)	0.0724** (0.0355)	0.0734** (0.0356)	0.167** (0.0659)	0.0650 (0.0444)
Shock _{it−1}	−0.0754** (0.0327)	−0.0670* (0.0355)	−0.0627* (0.0346)	−0.0645 (0.0661)	−0.0700* (0.0418)
Shock _{it−2}		0.0271 (0.0337)			
GDP growth _{it}	−0.125* (0.0730)	−0.0961 (0.0770)	−0.170** (0.0728)	−0.0991 (0.271)	−0.151* (0.0860)
USD xrate growth _{it}	−0.268*** (0.0788)	−0.261*** (0.0781)	−0.309*** (0.0798)	−0.544 (0.365)	−0.227*** (0.0765)
Non-U.S. import growth _{it}	0.353*** (0.0511)	0.369*** (0.0518)	0.348*** (0.0521)	0.494*** (0.180)	0.327*** (0.0544)
Observations	4949	4719	4798	709	4240
R ²	0.100	0.106	0.103	0.203	0.098

Note: This table reports results on the dynamic effects of trade finance supply shocks. The dependent variable is the growth rate of U.S. exports to country *i* at time *t*. *shock_{it}* is the constructed country-level trade finance supply shock. Columns (1)–(3) are estimated on the whole sample but with different lag and lead structures. Column (4) only includes observations during the crisis period from 2007 q3 to 2009 q2. Column (5) excludes these. All regressions include a constant, time- and country-fixed effects. Standard errors are bootstrapped and are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

letters-of-credit by a single bank mostly lead to a delay in export transactions until firms have found another bank that can provide the service, but they may harm trade in the long term if they occur during times of financial distress.³⁸

5.5. Separating the risk from the credit channel

Previous papers have focused largely on investigating how exports are affected by a contraction of credit through the working capital channel (e.g. [Amiti and Weinstein, 2011](#), [Paravisini et al., 2015](#)). In contrast, this paper is about the effect of reductions in the supply of letters of credit on exports, a channel we term the risk channel. There are three arguments that strongly favor this interpretation of our results.

First, as we discuss extensively in the data section of the Appendix, the trade finance claims in the FFIEC 009 data mostly reflect letters of credit. The fraction of the claims that cannot be accounted for by SWIFT letter-of-credit messages likely represents financing of trade between two non-U.S. countries but may also capture some pre-import financing for foreign importers of U.S. goods. Pre-import financing represents money lent by U.S. banks to foreign firms for the purpose of cash-in-advance.³⁹ A limited supply of pre-import financing implies a limited ability to settle a trade on cash-in-advance terms—hence a limited ability of U.S. exporters to eliminate the risk of non-payment by the importer—and can therefore be part of a wider definition of the risk channel.

While we are confident that our constructed country-level shocks indeed represent shocks to the supply of letters of credit, one may be worried that these shocks are correlated with the supply of loans at the bank level. If a firm obtained export loans from the same bank

that provides the guarantees, our estimates could also capture the working capital channel due to an omitted variable bias. To address this concern, column (4) of [Table 3](#) included as a control a measure of country-level loan supply as discussed in detail in [Section 5.1](#). Even though the bank-level shocks are correlated with bank-level loan growth as shown in [Table 2](#), country-level loan growth is not significant in the export regression.⁴⁰ In fact, the coefficient on the letter-of-credit supply shock is unaffected by its inclusion. This is strong evidence that we indeed identify effects on exports through the risk channel.

Finally, one should expect different effects from shocks to letters of credit than from shocks to loans. First, working capital needs are independent of payment risk, whereas the risk that the importer defaults determines whether an exporter demands a letter of credit. Moreover, working capital loans are fungible and firms can internally reallocate available funds. Letters of credit, in contrast, are destination specific and can only be obtained from a small number of banks. As our analysis shows, the shocks we identify only affect exports to some of the countries but not to others. More specifically we find that effects are driven by countries that are small and where few U.S. banks have operations. As small countries also tend to be riskier, these are exactly the countries we expect to be more affected by the risk channel.

6. Quantifying the effect of supply shocks

In this section, we conduct several experiments to explore the quantitative relevance of letter-of-credit supply shocks for U.S. exports. We will interpret the effect of γ as the direct effect of letter-of-credit supply shocks on export growth to country *i* in the narrative below. However, as discussed in [Section 4.1](#), γ probably represents an upper bound of this direct effect.

We start with the following experiment: we assume that a major letter-of-credit provider experiences a negative supply shock that corresponds to the 10th percentile of the bank shock distribution (a value of -0.426). Using this bank's market share in each destination country in the fourth quarter of 2011 and export values in the first and second quarters of 2012, the predicted aggregate effect on export growth is calculated as follows:

$$\Delta X_t = \frac{\sum_{i=1}^N (\gamma(-0.426)\phi_{bit-2}X_{it-1})}{X_{t-1}}. \quad (4)$$

We calculate ΔX_t for different values of γ . We start by setting γ equal to 0.0869, which corresponds to the estimated coefficient in column (3) of [Table 3](#). Based on this coefficient, such a letter-of-credit supply shock would reduce aggregate U.S. export growth by 1.3 percentage points. Taking into account the fact that reductions in the supply of letters of credit have larger effects for smaller destinations, this figure drops to 0.7% when applying the estimated coefficients in column (7) of [Table 5](#). When we assume that there is no effect for large export destinations and employ the coefficient in column 5 of [Table 5](#) to small markets, the effect on aggregate exports falls further to 0.1%.

In a next step, we demonstrate that it matters which bank is subject to the shock. We choose two large letter-of-credit suppliers and calculate the effect on export growth in selected regions of the world when each of them is hit by the shock described above. Columns (1) through (6) in the upper panel of [Table 7](#) show the results. Applying the coefficients from column (7) of [Table 5](#), exports to South Asia would fall by 0.94 percentage point if bank A were hit by the shock (see column (2) of [Table 7](#)), whereas the same relative reduction in

³⁸ [Alessandria et al. \(2010\)](#) argue that inventory dynamics were a key channel underlying the Great Trade Collapse. A fascinating question for future research is to which extent trade finance conditions and inventory dynamics interact and determine trade responses to shocks.

³⁹ As [Schmidt-Eisenlohr \(2013\)](#) and [Antràs and Foley \(2015\)](#) show, cash-in-advance terms tends to be used for the riskiest destinations. Cash-in-advance eliminates the risk that the exporter delivers but is not paid because the importer needs to pay for the goods before they are delivered.

⁴⁰ Note that country-level loan growth was constructed by using the same weights as for the construction of the country-level letter-of-credit supply shocks.

Table 7
Quantifications, in percent.

	Shock to bank A			Shock to bank B		
	Low	Med	High	Low	Med	High
	(1)	(2)	(3)	(4)	(5)	(6)
World	−0.11	−0.68	−1.34			
East Asia and Pacific	0.00	−0.58	−1.23	−0.77	−0.55	−0.46
Europe and Central Asia	−0.77	−0.98	−1.35	−0.91	−0.64	−0.52
South Asia	−0.17	−0.94	−1.82	−0.60	−0.45	−0.40
Sub-Saharan Africa	−0.23	−0.27	−0.37	−1.81	−2.12	−2.81
	Shock to all banks normal times			Shock to all banks crisis times		
	Low	Med	High	Low	Med	High
	(1)	(2)	(3)	(4)	(5)	(6)
World	−0.25	−1.09	−2.08	−0.44	−3.43	−4.38
East Asia and Pacific	−0.46	−1.19	−2.08	−0.79	−3.53	−4.25
Europe and Central Asia	−0.98	−1.41	−2.08	−1.67	−3.79	−4.25
South Asia	−0.50	−1.22	−2.13	−0.84	−3.63	−4.36
Sub-Saharan Africa	−1.17	−1.49	−2.06	−2.00	−3.86	−4.23

Note: The upper panel of the table shows the effect on export growth in different world regions if two different large banks in the U.S. were to reduce their supply of trade finance by a value of -0.426 , which corresponds to the 10th percentile of the bank-level shock distribution. The numbers shown in columns (1) and (4) are based on the shock coefficient in column (6) of Table 5, applied to small countries only. The coefficient applied to large countries is zero. Columns (2) and (5) are based on the shock coefficients from column (7) of the same table. Columns (3) and (6) use the baseline shock coefficient from column (3) of Table 3. The lower panel of the table displays the effect on export growth if all U.S. banks were subject to a moderate shock of -0.245 , which corresponds to the 25th percentile of the bank-level shock distribution. Columns (1) to (3) show effects during normal times, columns (4) to (6) apply the coefficients estimated for the 2007–2009 financial crisis. Column (1) to (3) are based on the same coefficients as in the upper panel of the table. Column (4) uses the coefficient from column (8) of Table 5. Column (5) uses the coefficients from column (9) of Table 5 and column (6) is based on the coefficient from column (1) of the same table.

letters of credit by bank B would reduce exports in this region by 0.45 percentage point (see column (5)). An even stronger asymmetry arises for Sub-Saharan Africa. This example illustrates that banks, through their global operations, can influence export patterns. The same bank shock affects countries differentially, depending on how important the bank is for the provision of letters of credit in each export market.

So far, we focused on what happens when only one of the banks reduces its supply of letters of credit. Next, we analyze the effect on exports if all banks were hit by a moderate shock that corresponds to the 25th percentile of the bank-level shock distribution (a value of $\alpha_{bt} = -0.245$) and roughly to half of the shock considered before. Depending on the loan growth coefficients applied, aggregate U.S. exports would fall between 0.25 and 2.1 percentage points (see the lower panel of Table 7). According to the results presented in Section 5.3, the effect of letter-of-credit supply shocks is larger during a crisis period. Effects would roughly double and range between 0.44 and 4.4 percentage points. Looking at the effect across world regions, Sub-Saharan Africa would be hit particularly hard by a reduction in the overall supply of letters of credit as the lower panel of Table 7 shows, due to its relatively large number of small U.S. export destinations.

As a final exercise, we compare the effect of a letter-of-credit supply shock to the effect of an exchange rate shock. According to the estimated coefficient in column (3) of Table 3, a 10-percent appreciation of the USD against the local currency of the importing country reduces U.S. exports by 2.52 percentage points. Hence, the effect of a negative letter-of-credit supply shock of one standard deviation during a crisis episode generates the same reduction in trade as an appreciation of the USD by 12.3%. The effects of letter-of-credit supply shocks are comparable to those of exchange rate changes.⁴¹

⁴¹ We estimate the contemporaneous exchange rate elasticity of U.S. exports. Estimates of the long-run elasticity are typically higher. See, for example Hooper et al. (2000). The beta coefficient of the exchange rate is 5.2%, comparable to the beta coefficient of letter-of-credit supply shocks, which is 5%.

The preceding exercises illustrate that letter-of-credit supply shocks are economically relevant for exports. This is particularly true for exports to small countries during times of financial stress, with a beta coefficient of 0.18. The overall role of trade finance for exports is likely larger. Recall that our estimation strategy mostly identifies effects of trade finance supply shocks through the risk channel. Shocks to the supply of letters of credit should affect trade flows over and above the effects of shocks to the supply of credit identified by Paravisini et al. (2015).

Note further that γ is the elasticity of export growth to bank-specific letter-of-credit supply changes. To disentangle the supply of from the demand for letters of credit, the estimated bank-level shocks are purged of any aggregate effects. While necessary for identification, it also means that aggregate letter-of-credit supply shocks, for example during the Great Trade Collapse, cannot be quantified and their effects cannot be estimated. This limitation is not specific to our paper but is a general feature of any cross-sectional estimation strategy like the ones employed by Amiti and Weinstein (2011) and Paravisini et al. (2015).

We have seen that reductions in the supply of letters of credit by single banks have larger effects during times of financial distress. This is probably because exporters find it harder to switch to other banks, which may be less willing to expand their balance sheets or to cooperate with new banks in foreign countries when uncertainty is high and liquidity limited. Switching might be even harder in the presence of an aggregate reduction in the supply of letters of credit, which, according to industry reports, happened in 2008. We therefore conclude that letters-of-credit very likely played a magnifying role in the Great Trade Collapse, affecting primarily exports to countries that are smaller and where fewer banks are active.

7. Conclusions

Exploiting data on the trade finance claims of U.S. banks that vary across countries and over time, this paper sheds new light on the effects of financial shocks on trade. While existing studies emphasize the working capital channel, this work provides evidence for the risk

channel based on new data that mostly capture letters of credit – a trade-specific, risk-reducing financial instrument. We show that shocks to the supply of letters of credit statistically and economically significant effects on exports.

While we follow the strategy of Greenstone et al. (2014) and Amiti and Weinstein (forthcoming) to identify supply shocks from the data, we modify and add new elements to the methodology. First, we obtain bank shocks separately for each country and show how to systematically drop information on similar countries to counter endogeneity concerns that may arise. Second, we demonstrate how sorting into markets can be excluded by jointly looking at serial correlation in bank shocks and by estimating the model using different lags of the market shares. These innovations can be useful for future empirical work.

Applying the approach, we find that exports to countries that are smaller, where fewer U.S. banks are active, are affected when banks reduce their supply of letters of credit. At the same time, changes in supply have stronger effects during times of financial distress. Another key result of the analysis is that single banks can affect exports in the aggregate. Due to the high concentration of the business, a large negative shock to one of the big U.S. letter-of-credit banks reduces aggregate exports by up to 1.3 percentage points. This effect more than doubles during times of financial distress. The presented findings suggest that letter-of-credit supply shocks can constrain exports to smaller destinations and during crisis episodes. Considering that reductions in the supply of letters of credit are associated with a contraction in bank lending and a rise in banks' credit default swap spreads, letters of credit may have a role in explaining the collapse in exports to the smaller countries in 2008–2009.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jinteco.2017.07.007>.

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